Course & TP Structure of System Administration

Class information:

- Start date: 21-Mar-26
- Study Time: 150 hours
- Time: 10h/Week
- Schedule: Saturday (8:00-12:00PM and 2:00-4:00 PM)
  Sunday (8:00-12:00PM)
- Room: F-107

System Administration Course is divided into three main sessions:

Session	Lecturer
Computer Networking	**KHEANG Hongly**
Operating System	**HENG Rathpisey**
System Administration	

System Administration Course Learning Outcome (CLO):

Session	CLO
Computer Networking	**CLO1:** Understand and Apply Basic Concepts of Computer Networking
	CLO2: Identify Network Hardware & Components
	CLO3: Understand Networking Model (OSI & TCP/IP)
	CLO4: Design and Define IP Address & Subnet in Network Environment
	CLO5: Practice and Apply Basic Configuration on Router and Switch
Operating System	**CLO6:** Understand Operating System Fundamentals and Linux Introduction.
	CLO7: Navigate Linux File System and Manage Files & Directories.
	CLO8: Manage Users, Groups, and File Permissions in Linux.
	CLO9: Understand Linux Process Management and System Monitoring.
	CLO10: Install and Manage Software Packages in Linux.
System Administration	**CLO11:** Install and Configure Linux Server Environment.
	CLO12: Manage Network Services and Configuration on Linux Server.
	CLO13: Configure and Manage Web Server and Database Services.
	CLO14: Implement Server Security and Firewall Management.
	CLO15: Automate System Administration Tasks and Backup Strategies.

Computer Networking – Detailed Course Learning Outcomes (CLOs):

- **Week-01**

  + Course Session

  - CLO1: Understand and Apply Basic Concepts of Computer Networking
  - Chapter Tittle: Networking Today
  - Chapter Material: Cisco (CCNA)
  - Chapter Outline:
    - Introduction to Computer Networking Concepts
    - Common Types of Computer Networking
    - Network Topology and Arrangement
    - Introduction to Network Security Concepts
    - Internet Connections and Internet Service Providers (ISP)

  + TP Session

  - Connect to Wireless Network Infrastructure (School or Lab Wi-Fi)
  - Establish Wireless Connection between Source and Destination
  - Setup Remote Access Connection on Windows Client over Local Network
  - Install Remote Control Application for Remote Access over the Internet
  - Setup File Sharing over Local Network on Windows Client
  - Learn and Setup File Sharing on Cloud Services over the Internet
  - Share Network Resources in Networking (Printer, Scanner, Projector and Camera)
  - Comparison Between Local Bandwidth and Internet Speed
  - Conclusion and Summary

- **Week-02**

  + Course Session

  - CLO2: Identify Network Hardware & Components
  - Chapter Tittle: Network Devices and Components
  - Chapter Material: Cisco (CCNA)
  - Chapter Outline:
    - Identify Network Devices and Describe Their Functionalities
    - Explain Network Connection Types (Wired & Wireless)
    - Describe Network Media Transmission

  + TP Session

  - Demonstrate about Network Devices and Components
  - Design Network Infrastructure and Topology using appropriate network devices
  - Apply Ethernet Patch Cable and Testing Tools

- Establish Wired Connection between Source and Destination
- Apply File Sharing over Network with P2P Topology (Ethernet Cable)
- Differentiate Between Wired and Wireless Bandwidth and Stability
- Conclusion and Summary

❖ **Week-03**

+ Course Session

- CLO3: Understand Networking Model (OSI & TCP/IP)
- Chapter Tittle: OSI & TCP/IP Model
- Chapter Material: Cisco (CCNA)
- Chapter Outline:
  - Introduction to Networking Model (OSI & TCP/IP)
  - Explain all 7 Layers and Protocols within Each Layers
  - Comparison OSI and TCP/IP
  - Describe Common Network Protocols in different Layers and their functionalities

+ TP Session

- IP address in Network Communication Concepts
- IPv4 Fundamentals and Binary Practice
- Identify IP address structure in Network ID and Host ID
- Define Network Size and IP Address Range Calculation

❖ **Week-04**

+ Course Session

- CLO4: Understanding IP Addressing & Subnetting in Network Environment
- Chapter Tittle: IP Address and Subnetting in Computer Networking
- Chapter Material: Cisco (CCNA)
- Chapter Outline:
  - Introduction to IP Address Concepts
  - IP Address Allocation and Internet Access
  - Differentiate Between Public and Private IP addresses
  - Network Segmentation and Subnetting Concepts

+ TP Session

- Identifying IP Addresses Classes in Public and Private Networking
- Subnetting Concept and Purpose Overviews
- Basic Subnetting in a Network Environment
- Design Subnetting (VLSM) that Meets the Given Requirements of Clients
- Assign Static IP address on Client Site and Starting Communication
- Design Different Networks to Meet the Requirements of Clients

- ❖ **Week-05**

+ Course Session

- CLO5: Practice and Apply Basic Configuration on Router and Switch
- Chapter title: Basic Router and Switch Management
- Chapter Material: Cisco (CCNA)
- Chapter Outline:
  - Introduction to Router and Switch Device
  - Data Routing Concept (Data Packet Forwarding)
  - NAT and PAT Operation on Router
  - Default Gateway on Router
  - Switching Concept (Frame Forwarding)
  - MAC address table on Switch
  - Different types of Switches and VLAN overviews

+TP Session

- Hardware Router and Switch Installation and Power Connectivity
- Access CLI to Router and Switch
- Apply Basic Configuration on Router and Switch
- Design and Configure Router and Switch that meet the requirements
- Test and verify the connection in network environments setup

Operating System – Detailed Course Learning Outcomes (CLOs):

- ❖ **Week-06**

+ Course Session

- CLO6: Understand Operating System Fundamentals and Linux Introduction
- Chapter Tittle: Introduction to Operating Systems & Linux
- Chapter Material: Linux Essential (LPI)
- Chapter Outline:
  - Introduction to Operating System Concepts and Functions
  - Types of Operating Systems (Windows, macOS, Linux, Mobile OS) History and Evolution of Linux and Open-Source Software
  - Linux Distributions Overview (Ubuntu, CentOS, Debian, Fedora) Linux Desktop vs. Server Environments
  - Introduction to the Linux Command Line Interface (CLI)

+ TP Session

- Install Linux Distribution on Virtual Machine (VirtualBox / VMware) Explore Linux Desktop Environment (GNOME / KDE)

- Access and navigate the Linux Terminal
- Execute Basic Linux Commands
- Understand Shell Types and Terminal Emulators Practice Command Syntax, Options, and Arguments Shut Down, Restart, and Log Out from Linux System

❖ **Week-07**

+ Course Session

- CLO7: Navigate Linux File System and Manage Files & Directories
- Chapter Tittle: Linux File System and File Management
- Chapter Material: Linux Essential (LPI)
- Chapter Outline:
  - Linux File System Hierarchy Standard (FHS)
  - Understanding Directory Structure
  - Absolute vs. Relative Paths
  - File and Directory Management Commands
  - File Viewing and Text Processing Commands
  - Wildcards and Globbing Patterns
  - Hard Links and Soft Links (Symbolic Links)

+ TP Session

- Navigate the Linux File System Using CLI
- Create, Copy, Move, and Remove Files & Directories (`touch`, `mkdir`, `cp`, `mv`, `rm`)
- View File Contents (`cat`, `head`, `tail`, `less`, `more`)
- Search for Files and Directories (`find`, `locate`, `which`)
- Use Wildcards for File Operations (`*`, `?`, `[]`)
- Create and Manage Hard Links and Symbolic Links
- Practice Text Filtering with `grep`, `wc`, and `sort`
- Conclusion and Summary

❖ **Week-08**

+ Course Session

- CLO8: Manage Users, Groups, and File Permissions in Linux
- Chapter Tittle: Users, Groups, and Permissions Management
- Chapter Material: Linux Essential (LPI)
- Chapter Outline:
  - Linux Multi-User System Concepts
  - User Account Types (Root, Regular, System Users) User and Group Management Commands

- Understanding Linux File Permissions (Read, Write, Execute) Permission Representation (Symbolic and Numeric/Octal) Special Permissions (SUID, SGID, Sticky Bit)
- File Ownership and Access Control

+ TP Session

- Manage System Services (Start, Stop, Configure Startup Type)
- Use Task Manager to Monitor and Manage Processes
- Create a System Backup Image
- Configure and Perform System Restore
- Configure Windows Update Settings
- Basic Troubleshooting using Event Viewer and System Logs

❖ Week-09

+ Course Session

- CLO9: Understand Linux Process Management and System Monitoring
- Chapter Tittle: Process Management and System Monitoring
- Chapter Material: Linux Essential (LPI)
- Chapter Outline:
  - Introduction to Linux Processes and Daemons Foreground and Background Processes Process States and Lifecycle
  - Process Monitoring and Management Commands Job Control and Signal Handling
  - System Resource Monitoring (CPU, Memory, Disk) Introduction to Systemd and Service Management

+ TP Session

- View Running Processes
- Manage Foreground and Background Processes
- Terminate and Send Signals to Processes
- Monitor System Resources
- Manage System Services with
- Schedule Tasks with Review System Logs

❖ Week-10

+ Course Session

- CLO10: Install and Manage Software Packages in Linux
- Chapter Tittle: Software Package Management in Linux
- Chapter Material: Linux Essential (LPI)
- Chapter Outline:
  - Introduction to Software Package Management Concepts
  - Package Managers in Linux (APT, YUM/DNF, Zypper)

- Repository Configuration and Management
- Installing, Updating, and Removing Packages Dependency Resolution and Package Information
- Compiling Software from Source Code
- Introduction to Snap and Flatpak

+ TP Session

- Update and Upgrade System Packages
- Search and Install Software Packages
- Remove and Purge Software Packages
- View Package Information and Dependencie
- Configure and Manage Software Repositories
- Install Software from Package Files
- Practice Compiling and Installing Software from Source
- Install and Manage Snap or Flatpak Packages

System Administration – Detailed Course Learning Outcomes (CLOs):

- **Week-11**

+ Course Session

- CLO11: Install and Configure Linux Server Environment
- Chapter Tittle: Linux Server Installation and Initial Configuration
- Chapter Material: Linux Server Administration
- Chapter Outline:
  - Introduction to Linux Server Administration Concepts
  - Differences Between Desktop and Server Linux Installations
  - Linux Server Installation (Ubuntu Server / CentOS / Rocky Linux)
  - Initial Server Configuration and Hardening
  - Remote Server Access with SSH (Secure Shell)
  - SSH Key-Based Authentication Setup
  - Introduction to Server Roles and Responsibilities

+ TP Session

- Install Linux Server Distribution on Virtual Machine (Minimal Installation)
- Perform Initial Server Setup (Hostname, Timezone, Network Configuration)
- Configure Static IP Address on Linux Server
- Install and Configure OpenSSH Server
- Connect to Linux Server Remotely via SSH Client (PuTTY / Terminal)
- Generate and Configure SSH Key Pair for Passwordless Authentication
- Disable Root Login and Password Authentication for SSH Security

- Introduction to Cloud Services and EC2 Overview
- Conclusion and Summary

❖ **Week-12**

+ Course Session

- CLO12: Manage Network Services and Configuration on Linux Server
- Chapter Tittle: Network Services and Configuration Management
- Chapter Material: Linux Server Administration
- Chapter Outline:
    - Linux Server Network Configuration and Management
    - Network Interface Configuration (Netplan / NetworkManager / nmcli)
    - DNS Server Concepts and Configuration (BIND)
    - DHCP Server Concepts and Configuration
    - Network Troubleshooting Tools and Techniques
    - Introduction to Network File Sharing (NFS & Samba)

+ TP Session

- Configure Network Interfaces Using Netplan or nmcli
- Verify Network Connectivity (ping, traceroute, netstat, ss, ip)
- Install and Configure DHCP Server (ISC DHCPD / dhcpd)
- Install and Configure DNS Server (BIND9) for Local Name Resolution
- Test DNS Resolution with dig and nslookup
- Setup NFS File Sharing Between Linux Server and Client
- Setup Samba File Sharing Between Linux Server and Windows Client
- Conclusion and Summary

❖ **Week-13**

+ Course Session

- CLO13: Configure and Manage Web Server and Database Services
- Chapter Tittle: Web Server and Database Server Administration
- Chapter Material: Linux Server Administration
- Chapter Outline:
    - Introduction to Web Server Concepts (HTTP/HTTPS)
    - Apache Web Server Installation and Configuration
    - Nginx Web Server Installation and Configuration
    - Virtual Host Configuration for Multiple Websites
    - SSL/TLS Certificate Configuration (Let's Encrypt)
    - Introduction to Database Server (MySQL / MariaDB)
    - Database Server Installation and Basic Administration

+ TP Session

- Install and Configure Apache Web Server
- Deploy a Sample Website on Apache

- Install and Configure Nginx Web Server
- Configure Virtual Hosts to Serve Multiple Websites
- Generate and Apply SSL/TLS Certificate for HTTPS
- Install and Configure MySQL / MariaDB Database Server
- Perform Basic Database Operations (Create Database, Create User, Grant Privileges)
- Connect Web Application to Database Server
- Conclusion and Summary

❖ **Week-14**

+ Course Session

- CLO14: Implement Server Security and Firewall Management
- Chapter Tittle: Linux Server Security and Firewall Configuration
- Chapter Material: Linux Server Administration
- Chapter Outline:
  - Introduction to Linux Server Security Best Practices
  - Firewall Concepts and Architecture
  - UFW (Uncomplicated Firewall) Configuration
  - Iptables Fundamentals and Rule Management
  - Fail2Ban for Intrusion Prevention
  - SELinux / AppArmor Security Modules Overview
  - System Auditing and Log Analysis

+ TP Session

- Configure UFW Firewall Rules (Allow, Deny, Limit)
- Enable and Manage UFW for SSH, HTTP, and HTTPS Services
- Understand and Write Basic Iptables Rules
- Install and Configure Fail2Ban to Protect SSH Access
- Review and Analyze System Logs (Journalctl, /var/log/messages)
- Monitor Failed Login Attempts and Blocked IPs
- Apply Security Hardening Checklist on Linux Server
- Conclusion and Summary

❖ **Week-15**

+ Course Session

- CLO15: Automate System Administration Tasks and Backup Strategies
- Chapter Tittle: Automation, Backup, and Disaster Recovery
- Chapter Material: Linux Server Administration
- Chapter Outline:
  - Introduction to Shell Scripting for Automation
  - Writing and Executing Bash Scripts
  - Task Scheduling with Cron Jobs and Systemd Timers
  - Backup Strategies (Full, Incremental, Differential)

    - Backup Tools in Linux (rsync, tar, dump/restore)
    - Introduction to Disaster Recovery Planning
    - Course Review and Final Assessment Overview

+ TP Session

- Write Basic Bash Scripts for System Administration Tasks
- Automate User Account Creation with Shell Script
- Schedule Automated Backups Using Cron Jobs
- Perform Full System Backup Using tar and rsync
- Create Disk Image Backup Using dd and partimage
- Restore Files and System from Backup using tar and dd
- Practice Disaster Recovery Scenario (Simulate Server Failure and Restore)
- Final Course Review, Q&A, and Exam Preparation
- Conclusion and Summary